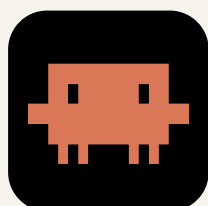


OPUS 5.5

Motion Design Prompting Techniques

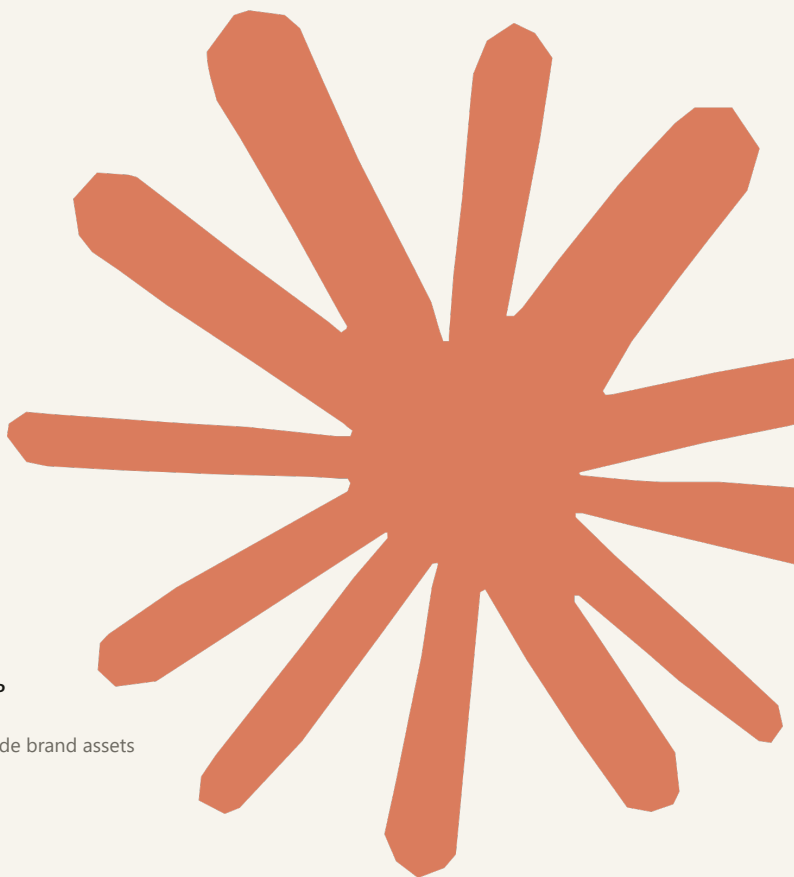
FROM ZERO TO MOTION DESIGN STUDIO

Director-grade prompts, deterministic rendering, motion systems, sound design and a critique loop that turns one-off clips into a repeatable studio pipeline.



BRIEF -> FRAMES -> SOUND -> REVIEW -> SHIP

Built from the supplied article, transcript and Claude brand assets





00 / MAP

Your studio route

Read cover to cover, or jump directly to the problem you need to solve. Every card is clickable.

01

FOUNDATION

Pages 03-05 / harness + prompt anatomy

02

DIRECTION

Pages 06-09 / reference + story + identity

03

PRODUCTION

Pages 10-12 / engine + motion + sound

04

PROMPT LIBRARY

Pages 13-16 / quick prompts + master briefs

05

REVIEW + SHIP

Pages 17-18 / self-critique + packaging the studio

Choose your first win

CALIBRATE

Start at 03, run the showreel prompt, verify rendering.

SELL

Start at 07, build a product film from real assets.

TEACH

Start at 15 or 16, then use the review gate on 17.

One rule for the entire guide

Plan first. Render evidence. Let Claude inspect its own frames before you call anything finished.



01 / REALITY

The prompt is 10%. The harness is 90%.

A one-liner may generate a clip. A studio pipeline makes quality repeatable.

10%

PROMPT

Genre, objective, constraints, reference

90%

HARNESS

Assets, seek(t), beat grid, frame review, export

01

BRIEF

A precise problem

02

SYSTEM

A deterministic renderer

03

EVIDENCE

Still, strips, phone test

04

LOOP

Score, fix, compare

MENTAL MODEL

Opus writes a program, not a video.

The program paints an exact frame for any time t . A browser captures frames. ffmpeg encodes them. The critique loop decides what gets fixed.



Install the studio, not just the model

The chat can design. Claude Code can render, listen, inspect frames and repair the result.

01

RUNTIME

Node 22+ / Python

02

CAPTURE

Playwright / Chromium

03

ENCODE

ffmpeg / H.264

04

OPTIONAL

Remotion / HyperFrames

COPY / CLAUDE.md

Render contract

- Every film is a pure function of time: `window.seek(t)`.
- No CSS transitions, timers or carried state in render mode.
- Seeded noise only. Never `Math.random`.
- Render H.264, yuv420p, CRF 16.

House look

- Ban centered title on gradient and "fade in everything".
- One display face, one UI face, one accent color.
- Something new must happen every 2-4 seconds.

Before delivery

1. Render one frame per beat as a contact sheet.
2. Score hook, phone readability, motion, variety, brand and sync.
3. Fix the 3 worst problems. Repeat until every score is 8+.

The rule that pays for everything else

Frame 812 must render correctly without simulating frames 0-811.



Write a director's brief

A strong prompt hires a virtual crew. Each block owns one kind of decision.

<role>

Who owns the film

<purpose>

What the viewer learns

<visual_identity>

Palette, type, texture

<story>

Scenes, VO, timestamps

<motion>

Transitions, camera, springs

<audio>

BPM, voice, tactile hits

<process>

Stills -> animatic -> review

<delivery>

Formats, poster, clean source

COPY / BRIEF SKELETON

You are the director, animator, sound designer and render engineer.

```
<purpose>Make [AUDIENCE] understand [ONE IDEA] and feel [EMOTION].</purpose>
<visual_identity>[LOOK], [PALETTE], [TYPE], [TEXTURE], [BANNED LOOKS].</visual_identity>
<story>Timed scenes. Every scene has visual, motion, VO, SFX and transition.</story>
<motion>No hard cuts. Visible objects physically become the next scene.</motion>
<audio>Measure or compose the beat grid before animation.</audio>
<process>Plan -> stills -> animatic -> polish -> self-critique.</process>
<delivery>MP4 + poster + contact sheet + source + README.</delivery>
```



A reference beats ten adjectives

Name the traits to transfer, and the content that must stay untouched.

CLAUDE CODE



```
01 <title>
02 Motion Design Director Prompt
03 </title>
04
05 <purpose>
06 Create a 45-second visual story
07 about [TOPIC] for [AUDIENCE].
08 Teach one idea, not a list.
09 </purpose>
10
11 <visual_identity>
12 Warm paper, coral accent,
13 charcoal type, tactile grain.
14 No gradients or generic glow.
15 </visual_identity>
16
17 <process>
18 1. Storyboard timed scenes.
19 2. Render a contact sheet.
20 3. Score every frame 1-10.
21 4. Fix the 3 worst issues.
22 </process>
23
24 <delivery>
25 MP4 + poster + clean source
26 </delivery>
```

VECTOR PROMPT SPECIMEN

Reference ladder

01

FRAME

Palette, grid, grain, type

02

VIDEO

Pacing, shots, transitions

03

LIBRARY

A visual DNA no one can copy

COPY / REFERENCE PASS

Reference: ./refs/launch.mp4 and ./refs/frames/*.png

1. Extract one frame every 0.5s and study the sequence.
2. Write docs/style_guide.md: palette, type, shot length, transition grammar, camera language, texture and grain.
3. Write docs/shotlist.md for a [DURATION] film about [SUBJECT].
4. Take the grammar of the reference - never its story, logo, characters, claims or copyrighted composition.
5. Show me both documents before writing animation code.

Take the grammar. Never take the content.



05 / BRAND FILM

Point the reel at a real product

Use actual UI, actual claims and actual type. Never redraw a product from memory.

The five-beat product story

01

PROBLEM

5 words

02

ASSEMBLE

real UI

03

PROVE

3 actions

04

METRIC

one number

05

LOCKUP

logo + CTA

COPY / PRODUCT FILM

Make a dynamic 20-second product film for [PRODUCT] ([URL]).

Assets

- Visit the site. Save real screenshots, logo, colors and fonts to ./assets.
- List every asset before animation. Never invent product UI or claims.

Story - one beat each, 2-4 seconds

1. Hook: the problem in 5 words of large kinetic type.
2. The product appears; the real UI assembles itself piece by piece.
3. Three features, each shown as a real action driven by a cursor.
4. One verified number that proves the value: [METRIC + SOURCE].
5. Logo lockup + [CTA].

Sound and output

- Original 120 BPM score. Clicks and whooshes land on measured peaks.
- Build 9:16 first, then recompose 1:1 and 16:9 from the same timeline.
- Show a contact sheet before the full render.

ASSET RULE

A brand film is credible only when the pixels are real.



06 / STORYBOARD

Give every beat a job

Style -> character -> transition -> scenes -> voice -> sound -> technical contract.

0-2s	2-8s	8-16s	16-26s	26-36s	36-48s	48-60s
HOOK	WORLD	QUESTION	MECHANISM	DISCOVERY	PAYOFF	LOOP
Impossible image	Visual rules	What to learn	Cause + effect	Experiment	Idea clicks	Return to start

COPY / SCENE CONTRACT

For every scene provide:

SCENE [N] / [START-END]

Learning purpose: the exact idea or product truth being communicated.

Visual: composition, characters, objects, labels and camera.

Motion: entrance, primary action, secondary motion and exit.

Voiceover: one sentence, written for 125-145 words per minute.

Sound: music cue, ambience and synchronized tactile effects.

Transition: the visible object that physically becomes the next scene.

No hard cuts. Do not duplicate narration as on-screen text.

Keep labels brief and readable at phone size.

Physical transitions carry meaning

sun rays -> black hole / confetti -> particles / sketch -> simulation

The viewer should witness one idea transforming into the next, not a deck changing slides.



Lock the world before it moves

A style block is not decoration. It is a constraint system for every scene and every collaborator.



01

PROPORTION

Head, body, silhouette

02

PALETTE

Sampled hex values

03

EXPRESSIONS

5 reusable states

04

MOTION

Idle, enter, react, exit

COPY / CHARACTER BIBLE

Create a character bible before animation.

- Identity lock: silhouette, proportions, face geometry and signature color.
- Expression sheet: neutral, curious, surprised, focused and delighted.
- Motion grammar: idle, point, inspect, react, celebrate and exit.
- Scale test: full body, waist-up, icon and 24 px silhouette.
- Background test: cream, charcoal and one accent field.

For every new style, preserve identity before adding texture.

Generate a single contact sheet and flag any panel that feels like a new character.

Visual identity block

WARM CREAM

CORAL

CHARCOAL

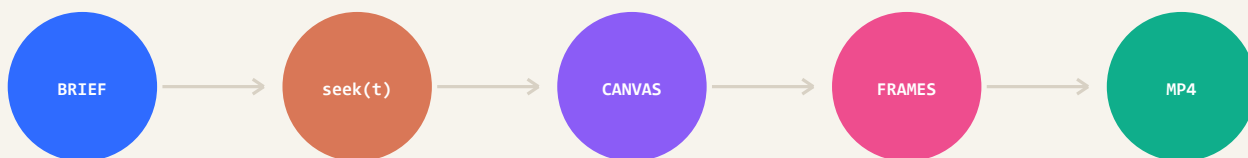
TEAL ACCENT

One display face. One UI face. One accent per scene. Texture belongs to the world, never on top of legibility.



Make time the only input

The renderer asks for frame t . Nothing depends on what happened one frame earlier.



MINIMAL seek(t) CONTRACT

```
<canvas id="c" width="1080" height="1920"></canvas>
<script>
const W = 1080, H = 1920, DUR = 15;
const g = document.getElementById('c').getContext('2d');

function draw(t) {
  g.fillStyle = '#171714'; g.fillRect(0, 0, W, H);
  // Every property is computed from t. No timers. No carried state.
  for (const scene of SCENES)
    if (t >= scene.from && t < scene.to) scene.draw(t - scene.from);
}

window.seek = (t) => { draw(t); return true; };
</script>
```

Render contract: 60 fps / 4 subframes / H.264 / yuv420p / CRF 16.

01

DETERMINISM

Render frame 300 twice. Hashes match.

02

MOTION BLUR

Blend subframes, not previous frames.

03

LOOP

Last state equals first, including velocity.



Give movement mass

Expensive motion accelerates, overshoots with intent and settles. It never feels like a PowerPoint slide.



SNAPPY

buttons / toggles

k 260 / d 24



DEFAULT

cards / camera

k 170 / d 26

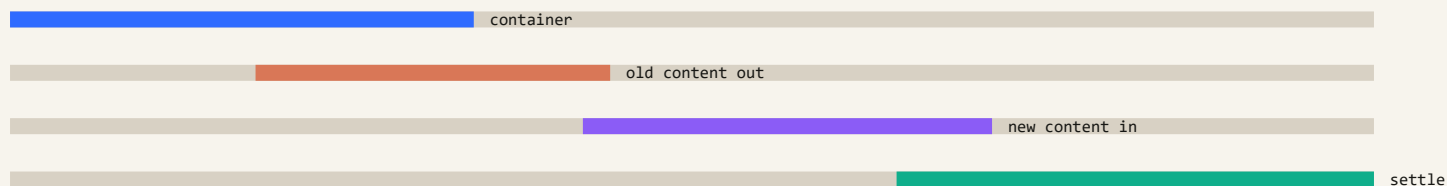


PLAYFUL

mascot / stickers

k 130 / d 18

Morph choreography



COPY / MOTION POLISH

Motion polish pass

- Replace every generic easing curve with a closed-form spring.
- Any value with multiple targets uses one additive spring per change.
- Text exits before the container morph; new text enters after it starts.
- Give leading and trailing edges different springs so indicators stretch.
- Tiny overshoot on UI, none on type, larger overshoot only for mascots.
- Hunt dead frames: every 2-4 seconds must introduce a new visual event.
- Render 12 consecutive frames around the fastest action and inspect for pops.

QUALITY TEST

If you can describe the motion only as slide, scale or fade, the scene needs a stronger physical idea.



Design sound before animation

The beat grid is a structural layer. Voice, cuts and tactile effects should agree on the same clock.



Voice + beat rules

- One voiceover line per scene, exported as a separate file.
- Write for 125-145 words per minute when teaching.
- Describe intention: warm, gentle, curious, precise.
- UI clicks land on peaks; hero reveals land on downbeats.
- Mix near -14 LUFS and keep narration clearly in front.

COPY / SOUND PASS

Sound pass

1. Analyze ./audio/track.wav. Write beats.json with BPM, beats, downbeats and transient peaks.
2. Snap state changes to beats and hero transitions to downbeats.
3. Generate one voice file per scene. Never time-stretch speech.
4. Create tactile SFX cues: click, pop, thump and whoosh.
5. Mix music, VO and SFX to -14 LUFS. Prevent clipping.
6. Re-render the contact sheet with timestamp + beat markers.

The voice key is VOICE_API_KEY in .env. Never print it.

Beat map



Small event every beat. New visual idea every 2-4 seconds. Hero reveal on a downbeat.



Six prompts. Six different jobs.

Copy, replace the brackets, then add the production gates from pages 04 and 17.

Showreel calibration

USE WHEN Testing the renderer

PROMPT

Make a dynamic 15-second motion graphics showreel. Every shot uses a different technique. Go all out.

WHY Genre knowledge supplies pacing and variety.

Anti-slop intro

USE WHEN You need a clean opener

PROMPT

Introduce [SUBJECT] in 10 seconds. Ban gradients, corner labels, frame borders, glow and fade-in-everything.

WHY Negative constraints remove the model's safest defaults.

Story over tricks

USE WHEN The reel has no idea

PROMPT

Use showreel energy to tell the history of [TOPIC] from [START] to today. 45 seconds, vertical.

WHY A timeline gives technique a narrative spine.

Product launch

USE WHEN Selling a real product

PROMPT

Make a 20-second film for [URL]. Use real screenshots, logo, colors, fonts and one verified metric.

WHY Real assets create trust and stop UI hallucination.

Reference translator

USE WHEN You have a style sample

PROMPT

Study ./refs. Write a style guide and shot grammar. Transfer the grammar, never the content.

WHY Separates visual language from copyrighted subject matter.

UI morph loop

USE WHEN Explaining many states

PROMPT

One container morphs through 8 UI states. A cursor drives every change. Last frame equals first.

WHY Continuity turns a feature list into one visual sentence.



Do not ask one prompt to do every job

Specialist passes make the output easier to inspect, compare and repair.

Style extractor

USE WHEN Before storyboarding

PROMPT

Extract palette, type, pacing, transitions, camera and texture from ./refs into style_guide.md.

WHY Creates a shared visual vocabulary.

Shotlist architect

USE WHEN Before animation

PROMPT

Turn the brief into timed shots. For each: purpose, visual, motion, camera, VO, SFX and transition.

WHY Makes omissions visible before they become code.

Motion director

USE WHEN After first render

PROMPT

Replace generic easing with springs. Inspect 12-frame strips around fast actions. Fix pops and dead frames.

WHY Targets physical credibility, not decoration.

Sound designer

USE WHEN Before final timing

PROMPT

Measure BPM and peaks. Put state changes on beats, reveals on downbeats and tactile SFX on actions.

WHY One clock makes the film feel intentional.

Harsh critic

USE WHEN Before delivery

PROMPT

Score the contact sheet 1-10. Name the 3 worst problems with timestamps. Fix and compare.

WHY Evidence replaces vague self-confidence.

Format director

USE WHEN Exporting a campaign

PROMPT

Recompose the same timeline for 9:16, 1:1 and 16:9. Preserve hierarchy; never crop the master.

WHY Responsive composition protects the idea.



13 / EDUCATIONAL FILM

Copy: storybook explainer

A complete 45-60-second director's brief. Replace the bracketed fields, then request the scene plan before code.

MASTER PROMPT / EDUCATIONAL

<role>

You are the director, educational storyteller, animator and sound designer.

</role>

<purpose>

Create a 45-60 second animation about [TOPIC] for [AUDIENCE].

Teach one core idea through a clear visual story, not a list of facts.

</purpose>

<visual_identity>

Paper cutouts, dry gouache, crayon and pencil texture. Torn edges with a thin white outline. Warm coral, mustard, teal, cream and charcoal. Slight stop-motion boil every few frames. One small mascot guides the viewer through every scene.

</visual_identity>

<story>

Write 7 timed scenes. For each: learning purpose, composition, objects, camera, motion, one VO sentence, sound cue and transition. Hook in the first 2 seconds. No hard cuts: a visible object in each scene physically becomes the next scene. Do not repeat narration as on-screen text. Keep labels short and readable.

</story>

<process>

1. Return the concept, visual system, character sheet and timestamped storyboard.
2. Build stills for all scenes and a contact sheet. Critique it at phone size.
3. Build a low-resolution animatic with placeholder VO. Fix pacing first.
4. Generate one VO file per scene; sync to the measured beat grid.
5. Render, score every criterion 1-10, fix the 3 worst issues, repeat 3 times.

</process>

<delivery>

1920x1080, 24 fps, captions, final.mp4, poster.png, contact.png and clean source.

</delivery>

Why it works

The style block builds a world. The mascot holds attention. The scene contract protects meaning. Physical transitions turn explanation into story.



14 / INTERACTIVE

Copy: sketch -> simulation

A friendly paper sketch becomes a real experiment with adjustable parameters and visible cause and effect.

2D SKETCH

friendly / paper / annotated



3D SIMULATION

play / measure / change / learn

MASTER PROMPT / INTERACTIVE

Build an interactive learning object about [SYSTEM / PHYSICS IDEA].

Opening state

- Begin as a hand-drawn paper sketch with labels and one clear question.
- The composition should feel like a notebook experiment, not a dashboard.

Transformation

- When the learner presses Play, the paper model unfolds into a real-time 3D simulation. **Make the 2D-to-3D transition the hero moment.**
- Simulate the underlying behavior; do not play a canned animation.

Controls

- Expose 3 meaningful parameters: [A], [B], [C]. Show units and safe ranges.
- Include Play, Pause, Reset and one suggested challenge.
- Update the result and explanation immediately when a control changes.

Learning design

- Ask for a prediction before each run, then compare it with the result.
- Highlight cause and effect with a trace, measurement and short insight.
- End with one transfer question: "What happens if...?"

Quality gate

- Test mouse, keyboard and phone layout. Keep 60 fps where possible.
- Explain model limits and assumptions. Deliver source + README.

Learning loop: predict -> play -> change one variable -> observe -> explain.



15 / CRITIQUE

Make Claude watch its own frames

Iteration is the method, not a failure of one-shot. Run at least three rounds. Nothing ships below 8/10.

HOOK / first 2s



PHONE READABILITY



MOTION QUALITY



VISUAL VARIETY



COMPOSITION



BRAND ACCURACY



SOUND SYNC



8+

or iterate

Contact sheet shows composition. Frame strip catches pops. Phone test proves readability. Loop check reveals the seam.

COPY / SELF-REVIEW

Open `contact.png`, `strip.png` and `phone.png`. Be a harsh motion director.

Score 1-10: hook, phone readability, motion, variety, composition, brand accuracy and sound sync.

List the 3 biggest problems with timestamps. Hunt for:

- text overlap during swaps or blurry scaled type
- linear slides, dead beats, generic centered titles and corner labels
- camera jumps, missing tactile SFX and a visible loop seam

Fix only the affected seconds. Re-render the sheets. Show old vs new scores. Repeat until every score is 8+ for three consecutive review passes.

Four proofs before export

CONTACT

STRIP

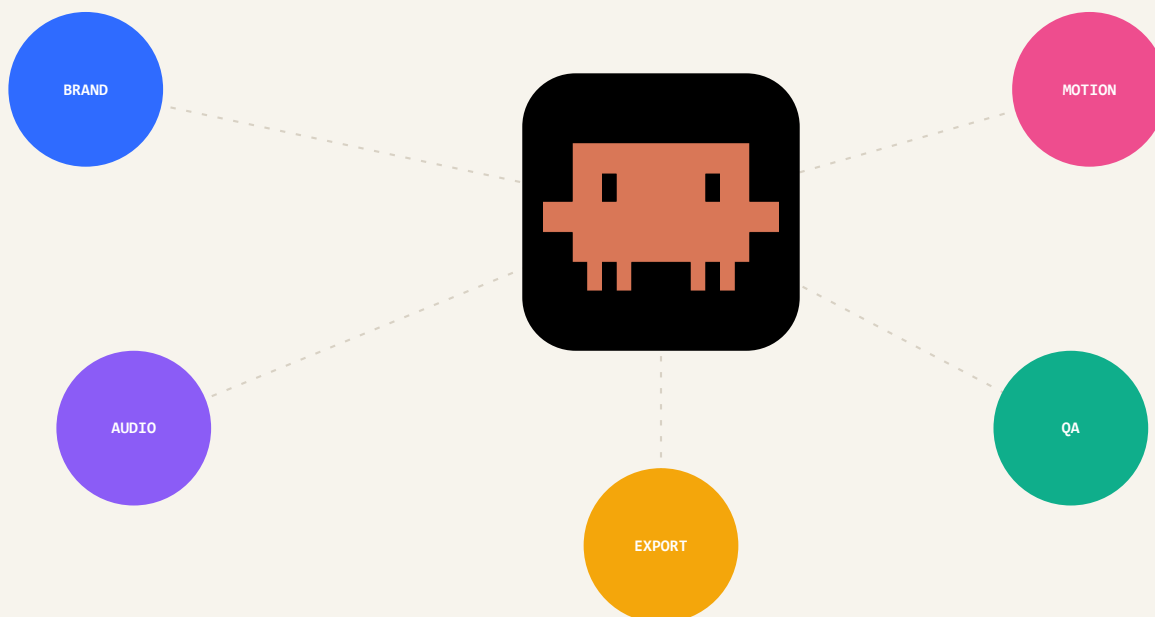
PHONE

LOOP



Package the pipeline

One timeline, multiple formats, one reusable skill and a delivery checklist the whole team can trust.



One layout-aware timeline

9:16

1:1

16:9

Recompose every format. Never crop the horizontal master.

ONE-LINE STUDIO COMMAND

```
/motion-reel for [URL], 20s,  
vertical, reference ./refs/frame.png
```

The skill loads a pipeline, not just a prompt.

SHIP CHECKLIST

- ☐ all final formats
- ☐ poster + contact sheet
- ☐ loop check
- ☐ clean source + README

The one-liner gets you a clip.
The harness gets you a studio.

SOURCE BASIS: USER-SUPPLIED ARTICLE + TRANSCRIPT + CLAUDE BRAND ASSETS